



Golden stars

We might have unequivocal evidence that gold and other heavy metals have come from the stars <https://digit.in/nov19-30>



Drone swarm exploration

Researchers have proposed using swarms of tiny drones to explore unknown environments <https://digit.in/nov19-29>

S

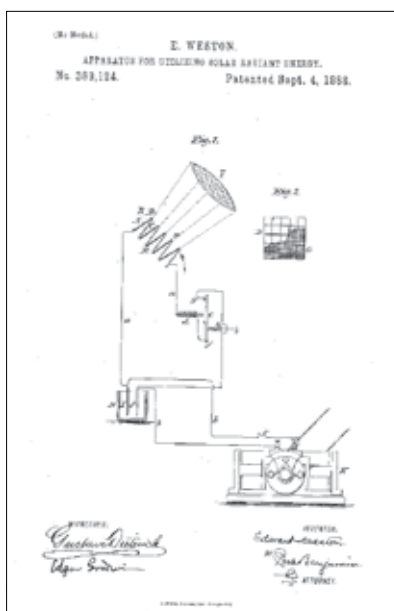
olar panels, as we know them today, are no more than 60 years old. However, the discoveries that

led to the creation of the first solar cell began 200 years ago. Discoveries about the properties of light and conductivity have transformed solar power into the behemoth alternative energy source it is today. Let's delve into the history of solar energy, rich with innovations and progress, that led to modern-day solar panels.

FIRST OBSERVATIONS AND DISCOVERIES

It all began with a young French physicist, Alexandre-Edmond Becquerel, observing and discovering the 'Photovoltaic effect' in 1839. It is the process where light or radiant energy is absorbed by a material which then creates an electrical current or voltage. A few decades after this, French mathematician, Augustin Mouchot, grew inspired by the aforementioned physicist's work and began registering patents in the 1860s for engines powered by solar energy.

In 1873, electrical engineer, Willoughby Smith unearthed the



"Apparatus for Utilizing Solar Radiant Energy", patented by Edward Weston on September 4, 1888

photoconductivity (becoming electrically conductive when absorbing light) of a material called selenium. Following this discovery, William Grylls Adams and Richard Evans Day discovered that selenium could output electricity from light without the need for any kind of heat or moving parts, which outlined the effectiveness of solar power since it was easy to harvest and maintain and required fewer parts than traditional energy sources.

Finally, in 1883, New York inventor, Charles Fritts, created the very first solar cell by layering selenium with a thin layer of gold. This archaic, but revolutionary, cell achieved an energy conversion rate of 1-2 per cent. In contrast, most modern-day solar cells achieve an efficiency rate of 15-20 per cent. After these radical discoveries were made, multiple inventors filed for patents for solar-powered devices as early as 1888.

In the year 1887, another advancement was made by German physicist, Heinrich Hertz, who first discovered the photoelectric effect. This is the

process where light is utilised to free electrons from a solid surface (usually metal) to generate power. In contrast to expecta-

SOLAR PANELS

Researchers have been advancing solar cell and panel technology for almost two centuries now

Dhriti Datta | dhriti@digit.in